

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Python for Data Science Practical

Practical - II

Roll No. : 38	Name: SIDDIQUE SAJID
Class : SY IT	Batch :
Date of Assignment :22-7-2026	Date/Time of Submission : 22-7-2026

Questions:

2a. Write a python code to create nested tuples.

code

```
student = ("sajid",38,("BSC IT","semester III"),90)
print("name =",student[0])
print("roll no =",student[1])
print("department =",student[2][0])
print("sem =",student[2][1])
print("marks =",student[-1])
```

output

```
==== RESTART: C:/Users/IT-03/Desktop/Python fo
name = sajid
roll no = 38
department = BSC IT
sem = semester III
marks = 90
```

2b. Write a python code to sort the nested tuple using sorted () function.

code

```
data = (3, "banana"),(1, "apple"),(2,"mango")
sorted_tuple = tuple(sorted(data))

print(sorted_tuple)
```

output

```
==== RESTART: C:/Users/IT-03/Desktop/Python fo
((1, 'apple'), (2, 'mango'), (3, 'banana'))
|
```

2c. Write a python code to copy or clone list.

```
original_list = [10, 20, 30, 40]
cloned_list = original_list.copy()

print(cloned_list)
```

output

```
----- RESTART: C.//
[10, 20, 30, 40]
```

2d. Write a python code to check immutability property of python tuples.

code

```
try:
    my_tuple = (10, 20, 30)
    my_tuple[0] = 99
except TypeError as e:
    print("Error:", e)
```

output

```
Error: 'tuple' object does not support item assignment
```

Curiosity Questions:

Q1. Define a tuple. How is it different from a list? Give one example.

Q2. Write a Python program to create the following tuples:

- a) An empty tuple
- b) A tuple with integers only
- c) A tuple with mixed data types
- d) A nested tuple (tuple inside another tuple)

Q3. Write a program to access elements of a tuple using indexing and slicing.

Q4. Create a tuple and find the following:

- Length of the tuple
- Maximum and minimum value (if all are numbers)
- Count of a specific item

Q5. Write a program to convert a list into a tuple.

Q6. Write a program to sort a tuple of numbers in ascending order.

Q7. Write a program to show that tuples are immutable (cannot be changed).

Q8. Write a program to clone (copy) a list and print both original and copied lists.

Q9. Write the output of the following code:

```
t = (1, 2, 3, 4, 5)
print(t[1:4])
```

Q10. Create a tuple of student marks: (78, 85, 92, 88, 76). Write a program to:

- Find average marks
- Display marks above 80